



Calculation Sheet

Calculation header

Identifier

*New calculation (1)***Medium selection and state**

Medium

Water

State

*Liquid***Operating data**☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	80.0	46.0	0.0	°C
p1	8.0	8.0	0.0	kgf/cm ² (g)
☐ p2	7.0	7.0		kgf/cm ² (g)
☒ Δp	1.0	1.0	0.0	kgf/cm ²
Cv	261.29	263.7		GPM(US)
☐ qm	220,680.0	224,760.0		kg/h
☒ qv	227.0	227.0	0.0	m ³ /h
LpAe	65.6	65.4		dB(A)
	Non-critical	Non-critical		

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ρ1	972.14	990.13	0.0	kg/m ³
ρ2	972.14	990.13		kg/m ³
☒ η1	0.35455	0.58577	0.0	mPa s
☐ ν1	0.36471	0.59161		mm ² /s
pv1	0.47414	0.10099		bar(a)
tv1	174.68	174.68	99.974	°C
cF1	1,556.0	1,539.2	0.0	m/s

Pipe downstream of valve

Size class downstream of valve

NPS2

2"

Schedule downstream

SCH2

STD



Calculation Sheet

Valve configuration

Valve type	<i>Rotary plug valve</i>
Trim type	<i>Eccentric plug valve</i>
Flow direction	<i>FTC</i>
Valve performance class	<i>Multi stage valve</i>
Protection	<i>Non-hardened</i>
Low-noise design	<i>n.a.</i>

Valve data

Basic characteristic		<i>Linear</i>	
☐ Lift stopper			
Selected valve size	NPS	<i>2"</i>	
Pressure class	class	<i>class 600</i>	
Maximum operating pressure	p,max	<i>105.0</i>	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL ²	<i>0.828</i>	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	<i>0.753</i>	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	<i>0.46</i>	-
Valve style modifier (Cv/Cv100 = 1)	Fd	<i>0.44</i>	-

Load-dependent values

	Maximum flow	Mean flow	Minimum flow	
u2	! <i>31.11</i>	! <i>31.11</i>		m/s
LpAe	<i>65.6</i>	<i>65.4</i>		dB(A)
Ri	! <i>1.4</i>	! <i>1.4</i>		-

Noise calculation

Calculation standard (noise, liquid)	<i>IEC 60534-8-4 (2015)</i>
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Noise prediction data

Low-noise design		<i>n.a.</i>	
Sound velocity	cF1	<i>1,556.0</i>	m/s
Pressure level	LA1	<i>154.0</i>	dB
Sound pressure level	Lpi	<i>154.0</i>	dB
Transmission loss	TL	<i>-76.8</i>	dB





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Noise prediction data (continued)

Sound power level (A-weighted)	LwAe	75.1	dB(A)
Sound pressure level of valve (A-weighted)	LpAe	65.6	dB(A)

Comments:

Reliability index - Ri

High velocity: 100 %
Recommended solution: Standard valve

Reliability index - Ri,op2

High velocity: 100 %
Recommended solution: Standard valve

Legend

- ✖ Errors
- ! Alarm

